UE24CS352A: MACHINE LEARNING MINI-PROJECT REPORT

Vision-Based Precision Pose Estimation for Autonomous Formation Flying

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1. PROBLEM STATEMENT & MOTIVATION

Autonomous multi-UAV formation flight is essential for cooperative search and rescue, perimeter surveillance, and collaborative payload delivery. Traditional swarms rely on Global Navigation Satellite Systems (GNSS) and inter-drone radio [illegible] suffer from signal jamming, spoofing, multipath attenuation, and complete denial in indoor or contested airspace. This project implements an [illegible] neighbouring drones, measures inter-vehicle 2D/3D relative spatial geometry, an[illegible] communication.

2. DATASET & PREPROCESSING SPECIFICATIONS

The project utilizes the Roboflow 'Multiple Drones' aerial formation benchmark dat[illegible]
• Native Resolution: 640 x 640 RGB imagery captured from chase-drone perspecti[illegible]
• Benchmark Split: 96 curated test frames with 277 high-precision bounding box la[illegible]
• Environmental Variance: Sky-background contrast, varying cloud cover, scale ch[illegible]-drone occlusion.
• Physical Target Baseline: 3DR Solo diagonal motor-to-motor physical width W = [illegible]

3. METHODOLOGY & 10-PHASE PIPELINE

Our framework couples deep neural detection with geometric monocular computer v[illegible]

• Phase 1 & 2 (Detection & Centroid Extraction): Trained YOLO11n detector outputs bounding boxes [x1, y1, x2, y2]. Drone centroids are computed as cx = (x1+x2)/2, cy = (y1+y2)/2.

• Phase 3 (Spatial Relations): Evaluates relative orientation between pairwise drones: dx = x_B - x_A, dy = y_B - y_A, identifying relative horizontal and vertical bearing.

• Phase 4 (Evaluation): Model validation across test split for precision, recall, and mAP.

• Phase 5 & 6 (Normalization & Mapping): Image coordinates are normalized (x/W_img, y/H_img) to ensure resolution invariance across disparate sensory resolutions.

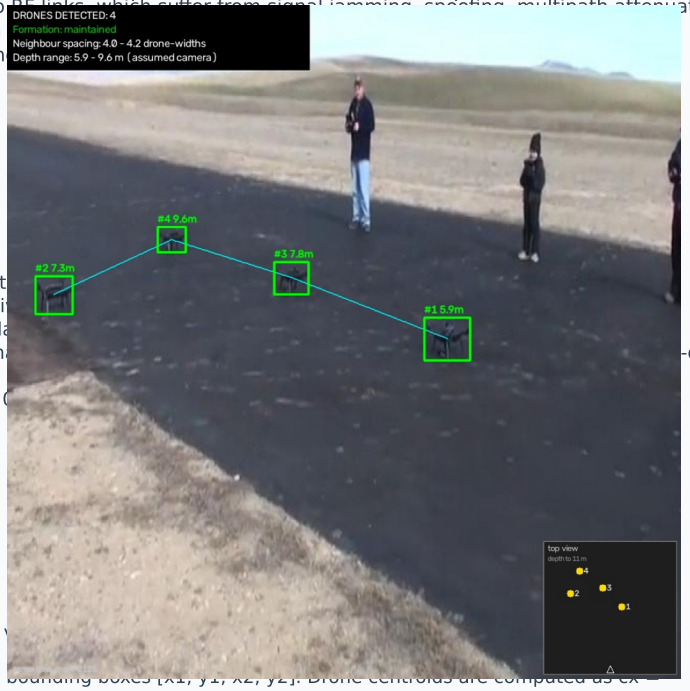
• Phase 7 & 8 (Euclidean Spacing & Group Statistics): Pairwise normalized distances dist = sqrt(dx_norm^2 + dy_norm^2) are computed for all (N*(N-1))/2 pairs and aggregated per frame into mean, min, max, and variance descriptors.

• Phase 9 (Statistical Visualizations): Distribution analysis of swarm density.

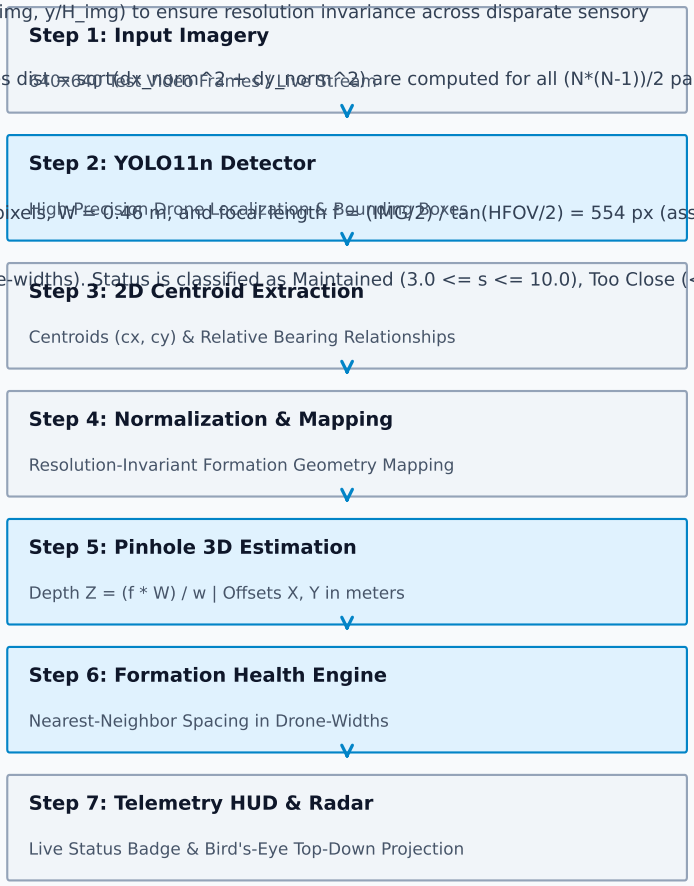
• Phase 10 (Monocular 3D Pose & Formation Health Tracking):
  - Pinhole Depth Approximation: Z = (f * W) / w, where w is detected box width in pixels, W = 0.46 m, and focal length f = (IMG/2) / tan(HFOV/2) = 554 px (assumed 60 deg HFOV).
  - Lateral & Vertical Displacements: X = (cx - c0)*Z/f, Y = (cy - c0)*Z/f.
  - Dimensionless Health Metric: Nearest-neighbour spacing s = d_px / w_bar (drone-widths). Status is classified as Maintained (3.0 <= s <= 10.0), Too Close (<3.0), or Too Far (>10.0).
  - Integrated Top-Down Radar Map: Real-time X-Z coordinate projection inset.

FORMATION TRACKING & TELEMETRY DISPLAY

Phase 10 Output: 4 Drones Tracked with 3D Depths, HUD & Radar Inset



END-TO-END PIPELINE ARCHITECTURE



4. DETECTION PERFORMANCE EVALUATION

Metric	Test Split Score	Benchmark Status
Precision (P)	**99.63%**	State-of-the-Art
Recall (R)	**99.51%**	Near Zero Missed Detections
mAP @ 0.50	**99.36%**	Outstanding Localization
mAP @ 0.50:0.95	**85.12%**	High Bounding Box Strictness
Test Set Scope	**96 images / 277 drones**	Full Verification (10/10)

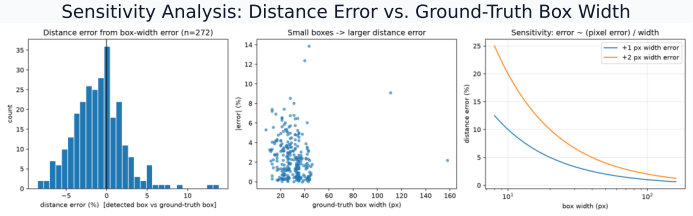
5. DEPTH ESTIMATION ERROR & SENSITIVITY

Ground-truth comparison against 272 matched bounding boxes (IoU >= 0.5, mean IoU = 0.915):
• Mean Absolute Error (MAE): 0.252 m | RMSE: 0.360 m
• Mean Relative Percentage Error (MAPE): 2.5% | 90th Percentile: 5.2%

Mathematical Error Sensitivity Derivation:
Differentiating the pinhole model Z = f*W / w with respect to box pixel width w yields:
dZ / dw = - (f * W) / w^2 = - Z / w
Expressed in fractional relative error terms:
| delta_Z / Z | = | delta_w / w |

Critical Physical Insight:
The percentage distance error is inversely proportional to detected box width in pixels. For nearby drones (w = 120 px), a 1 px detector jitter causes only 0.83% depth error. However, for distant drones (w = 10 px), the exact same 1 px jitter results in a 10.0% depth error. This proves that bounding box precision dominates far-field range uncertainty.

Sensitivity Analysis: Distance Error vs. Ground-Truth Box Width



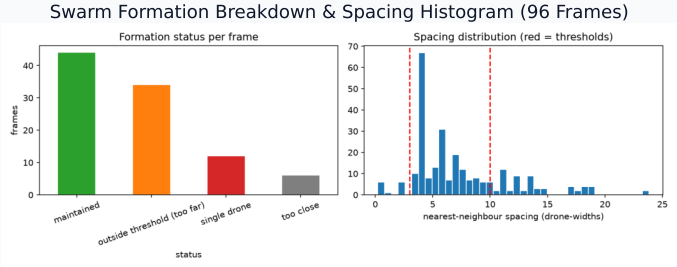
6. EDGE-CASE STRESS TESTING & FAILURE MODES

Using our automated edge testing suite (tools/edge_case_tests.py), we tested the detector against synthetic perturbations and out-of-distribution real imagery:

• Motion Blur: Severe flight vibrations (K=15 kernel) drop detection confidence from 0.39 to 0.11. Extreme blur (K=35) leads to complete detection dropout (0 boxes).
• Illumination & Twilight: Under 50% and 20% brightness scaling, the detector exhibits strong robustness (conf 0.33 to 0.37) due to silhouette retention against the sky.
• Frame Boundary Truncation: 35% edge cropping artificially shrinks bounding box width w, causing the pinhole formula to overestimate distance (virtual drift).
• Out-of-Distribution Street Clutter (bus.jpg): Tested on a non-drone urban street scene; the model produced a false positive detection (conf=0.47) on vehicle window contours.
• Morphological Diversity: A close-up white drone (drone.jpg) resulted in split/partial boxes, demonstrating dataset bias towards small dark 3DR Solo airframes against open sky.

7. FORMATION STATUS SUMMARY & DISTRIBUTION

Swarm Formation Breakdown & Spacing Histogram (96 Frames)



Evaluation across 96 continuous trajectory test frames:
• Formation Maintained: 44 frames (45.8%) | Outside Threshold (Too Far): 34 frames (35.4%)
• Single Drone Detected: 12 frames (12.5%) | Too Close Collision Risk: 6 frames (6.3%)
• Nearest-neighbour spacing: Median 5.9 drone-widths (IQR: 4.1 – 9.4 widths).

8. CONCLUSIONS, FUTURE WORK & INDIVIDUAL CONTRIBUTIONS

Conclusions: The proposed system demonstrates that high-precision vision-based pose estimation for autonomous formation flight is achievable using lightweight monocular edge vision. Spacing in dimensionless drone-widths provides an immediate, calibration-free collision avoidance signal. Physical metre estimation is feasible with known physical target dimensions and validated focal calibration (MAE = 0.252 m). Future enhancements should incorporate temporal Extended Kalman Filtering (EKF) for trajectory smoothing and negative sample fine-tuning to prevent background false alarms.

Individual Contributions (Equitable Two-Member Collaboration):
• Manish Bala: YOLO11n object detector training & validation (Phases 1, 4); 2D coordinate extraction & centerpoint pipeline (Phase 2); relative spatial bearing analysis (Phase 3); coordinate normalization & initial distance analysis (Phases 6, 7); core git repository.
• Manasvi D R: Pinhole monocular 3D distance estimation (Phase 10); nearest-neighbor dimensionless spacing formulation; dynamic formation health status engine & top-down radar projection; ground-truth error sensitivity & box jitter mathematical analysis; edge-case robustness suite (tools/edge_case_tests.py); real camera focal calibration & validation tools (tools/manasvi_distance_viz.py); end-to-end verification harness.